1 **(a)** 2 hours 35 minutes **[B1]**

(b) M1 for 15 35 or 17 70 seen

18 10 **[A1]**

[Total Marks: 3]

2 **(a)** 2 hours 15 minutes **[B1]**

(b) Adds 2 hours 35 minutes to 16 45 (e.g., 18 80) **[M1]**

19 20 **[A1]**

[Total Marks: 3]

3 Arrival time in London local time = 21 : 45 + 6 hours 50 minutes **[M1]**

04 : 35 (next day) **[M1]**

Dubai local time = 04 : 35 + 4 hours = 08 : 35 **[A1]**

[Total Marks: 3]

4 **(a)** London departure time = 10 30 or Dubai arrival time = 23 15 **[M1]**

8 hours 45 minutes **[A1]**

(b) London departure time = 18 45 or Dubai arrival time = 06 05 (Saturday) **[M1]**

02 05 Saturday **[A1]**

[Total Marks: 4]

5 **(a)** 22 50 + 14 h 35 min **[M1]**

13 25 **[A1]**

(b) 0.4×60 **[M1]**

3 h 24 min **[A1]**

[Total Marks: 4]

6 **(a)** Time from 09 45 to 10 00 = 15 minutes, and 10 00 to 11 20 = 1 hour 20 minutes **[M1]**

1 hour 35 minutes **[A1]**

(b) 14 55 + 4 hours = 18 55 **[M1]**

19 05 **[A1]**

[Total Marks: 4]

7 **(a)** 115 minutes = 1 hour 55 minutes **[M1]**

21 30 (or 21 : 30) **[A1]**

(b) Arrival time in New York = 21 40 + 7 h 15 m = 04 55 (next day) **[M1]**

Arrival time in Paris = 04 55 + 6 hours **[M1]**

10 55 (or 10 : 55) **[A1]**

[Total Marks: 5]

8 Flight arrival time in Dubai local time: 14 30 + 14 h 15 min = 28 45 (or 04 45 next day) **[M1]**

Adds time zone difference: 28 45 + 7 hours (or 04 45 + 7 hours) **[M1]**

11 45 **[A1]**

[Total Marks: 3]

9 (a) $3 \times 10^5 \times 60 \times 60 \times 24 \times 365$ **[M1]**

9.46×10^{12} or 9.4608×10^{12} **[A1]**

(b) $\dfrac{4.5 \times 10^{13}}{9.4608 \times 10^{12}}$ **[M1]**

4.76 **[A1]**

[Total Marks: 4]

10 (a) 11 : 03 − 08 : 15 **[M1]**

2 hours 48 minutes **[A1]**

(b) 48 minutes $= \dfrac{48}{60}$ hours = 0.8 hours **[M1]**

Average speed $= \dfrac{420}{2.8}$ **[M1]**

150 **[A1]**

[Total Marks: 5]

11 (a) M1 for arrival time in Dubai (10 00 Tuesday) OR departure time in London (22 15 Monday) seen

M1 for a valid time zone adjustment and flight duration addition

06 00, Tuesday **[A1]**

(b) M1 for $1.5 \times 365 \times 24 \times 60$

A1 for 788400

7.884×10^5 **[B1]**

[Total Marks: 6]

12 **(a)** London departure time = 16 30 − 14 h 15 min or 02 15 **[M1]**

Tokyo departure time = 02 15 + 9 h **[M1]**

11 15 **[A1]**

(b) 1 min 45 s = 105 seconds or 8 hours = 480 minutes **[B1]**

$(8 \times 3600) \div 105$ or $480 \div 1.75$ **[M1]**

274 **[A1]**

[Total Marks: 6]

13 **(a)** London departure time in Dubai local time = 13 45 + 4 hours = 17 45 **[M1]**

Time from 17 45 to 00 15 = (24 00 − 17 45) + 15 minutes = 6 hours 15 minutes +15 minutes **[M1]**

6 hours 30 minutes **[A1]**

(b) Flight time converted to decimal hours = 6.5 **[B1]**

Speed = $\dfrac{5486}{6.5}$ **[M1]**

844 **[A1]**

[Total Marks: 6]

14 **(a)** Time = $\dfrac{42.2}{12.5}$ or 3.376 hours **[M1]**

$0.376 \times 60 = 22.56$ minutes **[M1]**

$0.56 \times 60 = 33.6$ seconds **[M1]**

3 hours, 22 minutes, 34 seconds **[A1]**

(b) Correct difference calculation, e.g. 12 hours 64 minutes −08 hours 35 minutes **[M1]**

4 hours 29 minutes **[A1]**

[Total Marks: 6]

15 **(a)** M1 for $30 \times 24 \times 60$ or 43200 minutes seen

M1 for $\dfrac{43200}{94.5}$ or 457.142...

457 **[A1]**

(b) M1 for 100×94.5 or 9450 minutes

M1 for converting minutes into hours and days (e.g., 157.5 hours OR 6 days 13 hours 30 minutes)

M1 for adding their 6 days and 13 hours 30 minutes to 05 50 on 1st March

19 20, 7th March **[A1]**

[Total Marks: 7]

16 **(a)** Total minutes $= \frac{1\,000\,000}{80} = 12\,500$ **[M1]**

Total hours $= 12\,500 \div 60 = 208$ with remainder 20 (so 20 minutes) **[M1]**

Total days $= 208 \div 24 = 8$ with remainder 16 (so 16 hours) **[M1]**

8 days, 16 hours, 20 minutes **[A1]**

(b) Production start in Tokyo time = 08 00 + 14 hours = 22 00 on Monday **[M1]**

Add 8 days

$22\,00 on Tuesday (next week) [M1]$

Add 16 hours

$22\,00 + 16 hours = 38\,00$

$14\,00 on Wednesday [M1]$

Add 20 minutes

$Wednesday, 14\,20 [A1]$

[Total Marks: 7]

17 **(a)** $920 \div 24$ **[M1]**

38.333... hours **[A1]**

38 hours 20 minutes **[A1]**

(b) Adds their time from part (a) to Wednesday 17 50 **[M1]**

Wednesday 17 50 + 20 minutes = 18 10 **[M1]**

Wednesday 18 10 + 38 hours = Thursday 18 10 + 14 hours **[M1]**

Day: Friday **[A1]**

Time: 08 10 **[A1]**

[Total Marks: 7]

18 **(a)** $\frac{6600}{800}$ or 8.25 **[M1]**

8 hours 15 minutes **[A1]**

(b) Arrival time in City A time = 05 55 **[M1]**

00 55 Wednesday **[A1]**

(c) Departure from City B = 15 15 (local time) or 20 15 (City A time) **[M1]**

Arrival in City B time = 23 00 **[M1]**

04 00 Thursday **[A1]**

[Total Marks: 7]

19 **(a)** $\dfrac{35}{x} + \dfrac{42}{x-5}$ **[B1]**

(b) 4 hours 33 minutes $= 4 + \dfrac{33}{60}$ or $\dfrac{91}{20}$ hours **[B1]**

$\dfrac{35}{x} + \dfrac{42}{x-5} = \dfrac{91}{20}$ **[M1]**

Multiplies by common denominator: $20 \times 35(x-5) + 20 \times 42x = 91x(x-5)$ **[M1]**

$700x - 3500 + 840x = 91x^2 - 455x$ **[M1]**

$1540x - 3500 = 91x^2 - 455x \implies 91x^2 - 1995x + 3500 = 0$ **[M1]**

Divides by 7 to reach $13x^2 - 285x + 500 = 0$ without errors **[A1]**

(c) Uses quadratic formula or factors: $(13x - 25)(x - 20) = 0$ **[M1]**

$x = 20$ or $x = \dfrac{25}{13}$ (or 1.92...) **[A1]**

$x = 20$ only (rejects $\dfrac{25}{13}$ as speed $x - 5$ must be positive) **[A1]**

[Total Marks: 8]

20 **(a)** Minutes in a year $= 365 \times 24 \times 60$ **[M1]**

525600 minutes **[M1]**

Number of orbits $= \dfrac{525600}{92} = 5713.043\ldots$ **[M1]**

5713 **[A1]**

(b) **(i)** $\dfrac{11440}{880} = 13$ **[B1]**

(ii) Time difference $= 9 - (-4) = 13$ hours (Q is 13 hours ahead) **[M1]**

Departure time in City Q's local time $= 22:45$ (Monday) $+13$ hours $= 11:45$ (Tuesday) **[M1]**

Arrival time in City Q's local time $= 11:45$ (Tuesday) $+13$ hours (flight time) $= 00:45$ (Wednesday) **[A1]**

[Total Marks: 8]

21 **(a)** 15×92.75 or 1380 min 675 s **[M1]**

23 h 11 min 15 s **[A1]**

(b) Arrival time in City A local time $= 09\,45 + 8$ h 20 min or $18\,05$ **[M1]**

$26\,05 - 18\,05$ **[M1]**

8 **[A1]**

(c) Distance travelled by Train P at 08:30 $= 90 \times 0.5 = 45$ km **[M1]**

Relative speed of approaching trains $= 90 + 120 = 210$ km/h **[M1]**

Time taken to meet from 08:30 $= (360 - 45) \div 210 = 1.5$ hours **[M1]**

$10\,00$ **[A1]**

[Total Marks: 9]